Rahul Kota

Student Researcher

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About Me

I am a Pharm.D student driven by curiosity at the intersection of medicine, science, and technology. I am particularly interested in understanding biological systems at the molecular level and exploring how computational and data-driven approaches can contribute to biomedical research. My interests are expanding toward computational biology, bioinformatics, AI/ML, molecular biology, and biophysics, with a long-term interest in interdisciplinary research that connects healthcare sciences with fundamental biological mechanisms.

Education

Doctor of Pharmacy (III Year)

QIS College of Pharmacy

Affiliated to Governemnt Genral Hospital, Ongole, India and Jawaharlal Nehuru Technological University, Kakinada, India.

2024 - 2030

Intermediate (PCB Stream)

V.K.R College,
Gannavaram

CGPA : 9.02 **2021 - 2023**

Research Interests

- Breast Cancer
- Computational Oncology
- Molecular Biology
- Cancer Biology, Genomics
- Molecular Biophysics (Emerging Interest)

Languages

- English (Working Proficiency)
- Hindi (Working Proficiency)
- German A1 - (CEFR Level)

Research Experience & Internships

RESEARCH INTERNSHIP

Department of Chemistry,
National Institute of Technology Warangal (NITW)

Research Intern in the Dept of chemistry at NITW wroked in the Organic Research Laboratory. The lab focuses on Medicinal Chemistry, Drug Delivery and Gene Delivery.

GUIDED MENTORSHIP

Department of Pharmaceutical Engineering and Technology , Indian Institute of Technology (IIT BHU), Varanasi

Exploratory Cancer Biology & Nanocarriers Research
Literature-based exploratory research on miRNA-incorporated nanocarrier delivery for Triple-Negative Breast Cancer under PhD scholar guidance in the department(Remotely).

Projects & Publications

Research Article ·

TITLE: Source-Audited Prioritisation of miRNA–mRNA Regulatory Candidates in Triple-Negative Breast Cancer with Patient-Level Expression-Association Analysis
An integrative computational re-analysis of publicly available expression data to investigate miRNA–mRNA regulatory relationships in triple-negative breast cancer (TNBC).

Review Article ·

TITLE: *From marker–trait associations to functional genes: a bioinformatics framework for translating genomic variation into crop improvement: a review*
Conducted a bioinformatics-focused review of genomic variation analysis, functional gene prioritization, and marker–trait associations to explore how genetic discoveries can be translated into precision crop breeding and improved agronomic traits.

***Review Article* ·**

TITLE: Marine-derived extracellular vesicles as candidate nanocarriers: sources, cargo, therapeutic evidence and translational gaps

Critically reviewed marine-derived extracellular vesicles as emerging drug-delivery nanocarriers, with emphasis on oncology and doxorubicin delivery in triple-negative breast cancer (TNBC). Evaluated preclinical evidence for nanoalgosome-mediated anticancer activity and compared findings across cancer cell models and spheroids. Identified key translational gaps in tumour-targeting validation, therapeutic efficacy in animal models, cargo delivery, and safety.

Deployed Web App · 2025

AI-Assisted Healthcare Utility Website

Healthcare suite: adult & pediatric dose calculators, BMI assessment, and drug information - Python & Streamlit.
Web Link (Tap Here)

Peer Review & Editorial Contributions

Peer Reviewer - JMIR *(Journal of Medical Internet Research)*

Contributing to scholarly peer review by critically evaluating research manuscripts within relevant areas of my expertise. Focusing on scientific rigor, methodological quality, interpretation of findings, and clarity of research reporting.

Student Member

- *Student Member at American Association for Cancer Research* ***(AACR)***
  AACR ID - 2881450
- *Student Member at American Society of Clinical Oncology* ***(ASCO)***
- *Student Member at International Society of Oncology and Biomarkers* ***(ISOBM)***
- *Student Member - India Chapter at Society of Hematological Oncology* ***(SOHO)***

Workshops & Conferences

Conference · IISc Bangalore

5th Health Care Summit 2025

Attended the 5th Edition of Health Care Summit 2025 Held at Indian Institute of Science, Bangalore · June 2025

Workshop · IIT Indore

GIAN Course — Precision Medicines & Next-Gen Deep Medicines

Attended a 10 days Global Initiatve of Academic Networks (GIAN) workshop at Indian Institute of Technology Indore (IIT Indore), In collaboration with Purdue University, USA

Award · SCRMP · 2025

1st Place — Poster Presentation

Won First Prize in Poster Presentation during National Pharmacy Week 2025
Held by Society of Clinical Research and Medical Professionals (SCRMP), Gujarat Chapter

Tech Expo Project · QIS FEST 2026

Participated in the Technical Expo at QIS FE ST 2026,
Presented Smart RX AI, a software-based intelligent system designed to assist clinical decesion support and prescription analysis and Drug inteactions Using Computational Logic